# School of Computer Science and Artificial Intelligence

## Lab Assignment # 19.2

**Program : B. Tech (CSE)**

**Specialization :AIML**

**Course Title : AI Assisted Coding**

**Course Code : 23CS002PC304**

**Semester : VI**

**Academic Session : 2025-2026**

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**Batch No. : 33**

**Date :07/04/26**

## Task 1 – Python to Java Conversion

```python
with open("input.txt", "r") as file:
    data = file.read()

with open("output.txt", "w") as file:
    file.write(data.upper())

print("File processed successfully")
```

## Prompt:

Convert this Python file handling code to Java with same functionality and proper exception handling.

## Code:

```java
public class FileHandlingExample {
    public static void main(String[] args) {
        try {
            BufferedReader reader = new BufferedReader(new FileReader("input.t
            BufferedWriter writer = new BufferedWriter(new FileWriter("output.

            String line;
            while ((line = reader.readLine()) != null) {
                writer.write(line.toUpperCase());
                writer.newLine();
            }

            reader.close();
            writer.close();

            System.out.println("File processed successfully");

        } catch (IOException e) {
            e.printStackTrace();
        }
    }
}
```

## Task 2 – C++ to Python (Sorting Algorithm)

```cpp
#include <iostream>
using namespace std;

int main() {
    int arr[5] = {5, 2, 9, 1, 3};
    int n = 5;

    for(int i = 0; i < n-1; i++) {
        for(int j = 0; j < n-i-1; j++) {
            if(arr[j] > arr[j+1]) {
                swap(arr[j], arr[j+1]);
            }
        }
    }

    for(int i = 0; i < n; i++) {
        cout << arr[i] << " ";
    }
```

**Prompt:**

```
Convert this C++ sorting program into Python with same logic and output.
```

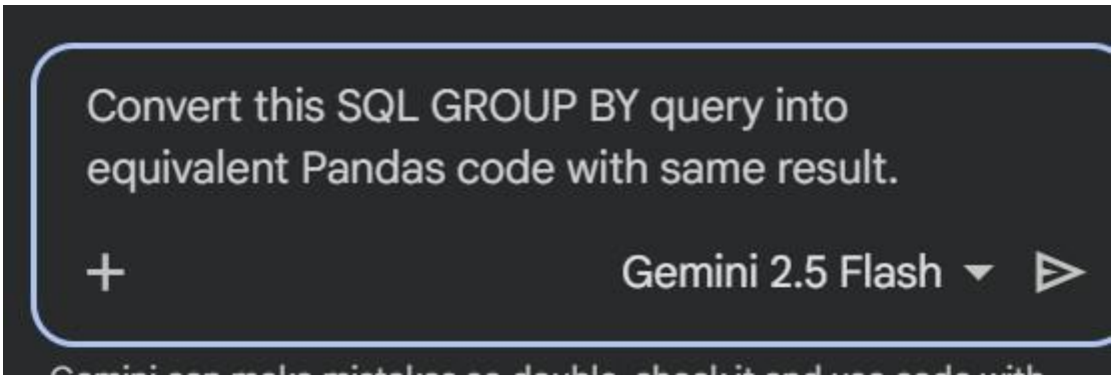
**Code:**

```python
arr = [5, 2, 9, 1, 3]

n = len(arr)

for i in range(n-1):
    for j in range(n-i-1):
        if arr[j] > arr[j+1]:
            arr[j], arr[j+1] = arr[j+1], arr[j]

print(arr)
```

**Task 3 – SQL to Pandas Query**

```sql
SELECT department, SUM(sales) as total_sales
FROM company
GROUP BY department;
```

**Prompt:**



**Code:**

```python
import pandas as pd

data = {
    'department': ['A', 'B', 'A', 'B', 'C'],
    'sales': [100, 200, 150, 300, 250]
}

df = pd.DataFrame(data)

result = df.groupby('department')['sales'].sum()

print(result)
```

**Task 4 – Real-Time Application (Python Flask → Java Spring Boot)**

```python
from flask import Flask

app = Flask(__name__)

@app.route('/')
def home():
    return "Hello World"

if __name__ == "__main__":
    app.run()
```

**Prompt:**

Convert this code from one language to another and compare both implementations.

**Code:**

```
import org.springframework.web.bind.annotation.*;

@RestController
public class HelloController {

    @GetMapping("/")
    public String home() {
        return "Hello World";
    }
}
```

**Task 5 – Python to C++ (Even/Odd Check)**

```
def check_even_odd(num):
    if num % 2 == 0:
        print("Even")
    else:
        print("Odd")

check_even_odd(5)
```

**Prompt:**

```
Convert this Python even/odd function into C++ and test with inputs.
```

**Code:**

```cpp
#include <iostream>
using namespace std;

int main() {
    int num;
    cin >> num;

    if(num % 2 == 0)
        cout << "Even";
    else
        cout << "Odd";

    return 0;
}
```